
Empowering Agents in Open-Ended Settings

Catherine Ji[†]

Department of Physics
Princeton University
cj7280@princeton.edu

Vivek Myers[†]

Department of Computer Science
UC Berkeley
vmyers@berkeley.edu

Sergey Levine

Department of Computer Science
UC Berkeley

Benjamin Eysenbach

Department of Computer Science
Princeton University

August 15, 2026

Abstract

Empowerment measures the capacity of an agent to actively control its environment. This object has been identified as an unsupervised mechanism for natural and artificial agents to make sense of large, open-ended environments. In this work, we provide arguably the first demonstration of empowerment maximization in such open-ended, large environments. In addition to achieving state-of-the-art performance in standard exploration metrics, empowered agents perturb their environment in interpretable ways. Our work supports empowerment maximization as a principled mechanism to train open-ended agents. Anonymized codebase and videos are at this link: [link].

1 Introduction

In large state spaces, searching for sparse rewards is a hard, often impossible task. An undirected agent often cannot stumble upon interesting behavior using random actions. However, real environments have inherent structure that makes the problem of searching for sparse reward more tractable. For instance, an agent may wish to accumulate resources and gain control over certain objects in an environment, so these objects may serve as a resource for arbitrary future task; most interesting tasks in an environment (e.g. via externally-defined rewards) often align with the same degrees of freedom as resource accumulation. In this work, we introduce a method to train agents that maximizes the generic, universal resource of *control*: we reward an agent for its ability to exert control over an environment. We quantify control using the information-theoretic quantity of **empowerment**, and build on state-of-the-art skill learning methods to maximize empowerment in open-ended settings.

Empowerment measures the degree to which an agent can control their environment. Prior literature has hypothesized and shown that empowerment (and similar objects) maximization can help explain how natural and artificial agents build causal models of the world and interact with open-ended environments. [cite Alison’s, Dave’s work, Alex Lewandowski’s, Friston kind of work, etc.]. For example, children [at what age? need to check paper] behavior in open-ended game settings is better described by empowerment, instead of alternative intrinsic motivations like curiosity or information gain [Gopnik paper, children playing Crafter]. Indeed, empowerment speaks to a general need for RL methods that are compatible with the Big World Hypothesis: real-world agents should successfully interact with an environment that they cannot fully describe. Despite a long line of literature studying the behavior of empowerment-maximizing agents, there has not yet been empirical demonstrations of empowerment

[†]Equal contribution.

maximization in such challenging, open-ended settings that escape simple characterization. Early literature (VIC [1] and its predecessor) estimate and maximize empowerment *as a channel capacity* using variational methods [2–4]. However, most of these works only scale up to simple tabular settings or low dimensional continuous settings. These methods eventually gave rise to a later line of unsupervised skill discovery literature successful in higher-dimensional locomotion and manipulation tasks.

We aim to close this gap between concept and current practice. Our starting observation is that state-of-the-art mutual information skill learning methods already maximize quantities similar to empowerment; MISL maximizes value functions that lower bound empowerment. In this work, we present a recipe to adapt unsupervised skill learning methods to maximize empowerment in challenging, high-dimensional settings. Along the lines of [scaling up option learning paper from Michael Matthews], the goal of this work is to provide methods to tractably train empowerment-maximizing agents. Using SOTA skill learning methods, we present SOTA exploration results in Craftax that beat previous exploration baselines and performance of prior empowerment maximization methods (adapted to the high dimensional and/or continuous setting). Furthermore, our empirical results show that empowerment maximization rewards behavior distinct from canonical intrinsic exploratory methods, prioritizing resource and skill accumulation over novelty. Ablations reveal the impact of different implementation decisions in large discrete and continuous environments that support theoretical results in prior works on discrepancies between information geometries and reward geometries.

Our work suggests that empowerment maximization is a tractable mechanism for agents to explore large, open-ended settings. We present future directions to make empowerment maximization methods more robust and scalable, particularly in the continuous setting.

2 Related Work

Intrinsically-Motivated Exploration Methods.

- Intrinsically motivated methods are exploration methods in RL that require no external information about the Markov Decision Process (MDP), outside of environment interaction. Introduced to be useful in the sparse reward setting, where the agent must navigate and figure out a big, complicated world without much signal.
- Roughly categorized into curiosity, novelty, information gain, prediction error (cite the psych paper). Typical methods optimize these intrinsic rewards alongside extrinsic rewards.
- Scalable methods include RND (prediction error to output of a randomly-initialized neural network), E3B (ellipticity metric that rewards exploration in low-variance directions), BYOL-Explore, temporal distances (e.g. Go-Explore (hard-to-reach goals), ETD, C-TeC).
- Also an entire suite of proposed exploration methods that have strong theoretical properties, but challenging to optimize in high-dimensional settings – often the very open-ended settings that would benefit from exploration beyond extrinsic rewards. E.g. novelty over the state occupancy, Lewandowski paper, (see Reward... paper w/ the local linearization argument). Empowerment is one of these objects (cite Dave’s paper).
- Despite the promise of these methods, including standard objectives like RND, often show no benefits in benchmarks we may care about. There are immediately some problems – these objectives are often engineered to decay under repeated exposure to the same observations, and are by-definition non-stationary which, when combined with bootstrapping and function approximation, becomes problematic (e.g. some of these issues addressed in RND paper – for instance, separate value functions for intrinsic versus extrinsic rewards).
- In this work, we do not purport to solve these problems that are generically true of intrinsic rewards, and focus on the problem setting optimizing *the intrinsic empowerment signal independent of extrinsic signal*. Not a new problem setting (see CTeC paper, skill discovery paper, etc.).

Unsupervised Skill Discovery. Unsupervised skill discovery methods learn latent-conditioned policies without access to extrinsic (supervised) reward signals. [CJ: add cites] Mutual Information Skill Learning (MISL) forms a subset of these methods, learning policies that maximize particular mutual information (MI) between observations and actions [5]. Generally, MISL methods learn policies such that latents are informative of future states [CJ: we definitely need a figure here to provide addnl information].

Potential and Effective Empowerment.

- As originally defined, potential empowerment is the channel capacity at a particular state, quantifying how much an agent can control from a particular state. (cite Capdepuy and our prior paper). Repeat a lot of info from the potential and effective empowerment section. Effective empowerment measure how many active bits...
- In works in psychology and the cognitive sciences, empowerment has been conjectured to explain human behavior in complicated/big settings not captured by other intrinsic motivations like curiosity [6–9] – for example, an agent seeking to control the world would not choose to stare at a TV of noisy pixels (e.g. the Noisy TV problem) which particularly affects (probably RND?). Desirable behavior for a human, but also for assistance problems [10, 11].
- Other works have conjectured empowerment and related objects as suitable objectives to train artificial intelligent agents (Dave’s and related papers, Mohamed + Rezende).
- Baselines: VIC [1] (implemented), VIC revisited paper. Cite Levy’s paper [12], though not scaled.
- Underlying URL methods: CSF/METRA, DIAYN, GenDa

Skill Learning and Effective Empowerment.

- MISL objectives have a close relationship to empowerment
- Importantly, the effective empowerment object and the skill learning objects are similar, even though the most recent modern skill learning methods do not reference empowerment. Indeed, the predecessor to modern MISL methods (Mohamed + Rezende + VIC [1, 2]) explicitly aim to maximize forms of empowerment.
- Specifically, modern MISL methods optimize effective empowerment – learning skills that actively transmit bits to the future.
- Effective empowerment lower bounds potential empowerment, as pointed out in (Levy paper, maybe old Mohamed + Rezende?). However, these empowerment papers did not scale to continuous/high dim settings.
- In this work, we aim to close the loop and optimize empowerment using modern MISL methods + empirical demonstrations in large, open-ended settings – the setting where empowerment matters. We introduce methods + release on and off-policy implementations, highlighting some key considerations when optimizing empowerment practically: (1) empowerment over observation vs. state space, (2) making empowerment well-defined in continuous spaces, and (3) breaking symmetry in empowerment policies. Strikingly, empowerment (maxes out? does very well on?) Craftax-Classic, a etc.
- The key contribution of our paper is practically adapting modern ... to empowerment maximization, presenting what is arguably the first demonstration of empowerment maximization in a large, open-ended setting. Our work opens up empowerment maximization as a tractable objective, pushing forward research on intrinsically-motivated agents in open-ended settings.

3 Preliminaries

MDP problem setting. We apply our method to fully-observed MDPs with states $s \in \mathcal{S}$, actions $a \in \mathcal{A}$, and environment dynamics $p(s' | s, a)$. Let the skill-conditioned policy be the function $\pi(a | s, z)$ with skill $z \in \mathcal{Z}$.

The latent skill z lives in a skill space $\mathcal{Z}$, which may be discrete or continuous. In the case of discrete skills, the set $\mathcal{Z} = \{1, 2, \dots, M\}$ is a finite set of M skills where z indexes a family of M policies. In the case of continuous skills, the latent $\mathcal{Z} \subseteq \mathbb{R}^d$ lies within a continuous latent space (i.e. the surface of a hypersphere). Given a skill z , the skill-conditioned policy $\pi(a | s, z)$ is a distribution over actions given the current state s , the latent z , and the start state s_0 . We write $\Pi_{\mathcal{Z}}$ for this class of skill- and start-state conditioned policies. The start-state s is the context of a skill [13].

Policies and value functions. Alongside $\Pi_{\mathcal{Z}}$, we write Π for the policies $\pi(a | s)$ that do not condition on a skill. Fix a discount factor $\gamma \in (0, 1)$. For a reward $r : \mathcal{S} \rightarrow \mathbb{R}$, the value function of a policy $\pi \in \Pi$ is the discounted return that the policy collects from a state,

$$V_r^\pi(s) \triangleq \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r(s_t) \mid s_0 = s \right], \quad (1)$$

where the expectation is over trajectories with actions $a_t \sim \pi(a | s_t)$ and states $s_{t+1} \sim p(s' | s_t, a_t)$. Two rewards appear in this work. The potential policy π_{pot} takes the potential empowerment as a reward (Definition 3.2). The skills π_{eff} take the reward supplied by the MISL objective, with skill value functions $V_z^\pi(s; s_0)$ defined in Appendix A.

Discounted state occupancy. For some initial state s_0 , skill latent $z \in \mathcal{Z}$, and skill $\pi(a | s, z)$, the discounted state occupancy measure (DSOM) takes the form

$$p^\pi(s_+ = s | s_0, z) \triangleq (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t p^\pi(s_t = s | s_0, z), \quad (2)$$

where $p^\pi(s_t = s | s_0, z)$ is the probability that policy π visits state s after t time steps when initialized at state s_0 under skill z .

We write the on-policy, skill-conditioned DSOM as $p^\pi(s_+ | s_0, z)$. Averaging over the skill distribution $p(z | s_0)$ gives the skill-marginal DSOM,

$$p^\pi(s_+ | s_0) = \mathbb{E}_{p(z | s_0)} [p^\pi(s_+ | s_0, z)], \quad (3)$$

the state distribution induced by the policy when the skill is unknown.

Potential and Effective Empowerment. Our work highlights the difference between empowerment, as originally defined in Klyubin et al. [14] and the MI objects directly maximized in modern MISL methods. We respectively term these forms of empowerment as potential and effective empowerment. Prior works have pointed out this distinction between these forms of empowerment [15].

The definitions are as follows:

Definition 3.1 (Potential empowerment). Let policies $\pi \in \Pi_{\mathcal{Z}}$ condition on skill latents $z \in \mathcal{Z}$ and let S_+ denote the random state $s_+ \sim p^\pi(s_+ | s_0, z)$ sampled from the DSOM. Potential empowerment is the channel capacity over a *family of policies* dependent on an initial state [3, 15]

$$\mathcal{E}_{\text{pot}}(s_0) = \max_{\pi \in \Pi_{\mathcal{Z}}} I^\pi(Z; S_+ | S_0 = s_0).$$

The definition of a potential empowerment maximizing policy follows:

Definition 3.2 (Potential empowerment policy). A potential empowerment policy $\pi_{\text{pot}} \in \Pi$ takes the potential empowerment of Definition 3.1 as a reward. Setting $r = \mathcal{E}_{\text{pot}}$ in Eq. (1) gives the value of a policy $\pi \in \Pi$ at a state s ,

$$V_{\mathcal{E}_{\text{pot}}}^{\pi}(s) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t \mathcal{E}_{\text{pot}}(s_t) \mid s_0 = s \right], \quad (4)$$

and π_{pot} maximizes $V_{\mathcal{E}_{\text{pot}}}^{\pi}$ at every state,

$$\pi_{\text{pot}} \in \arg \max_{\pi \in \Pi} V_{\mathcal{E}_{\text{pot}}}^{\pi}(s) \quad \text{for all } s \in \mathcal{S}. \quad (5)$$

While conceptually similar, mutual information skill learning methods do not maximize potential empowerment, and maximize effective empowerment. *Effective empowerment* captures the MI itself:

Definition 3.3 (Effective empowerment). Consider skill and start state-conditioned policies $\pi \in \Pi_{\mathcal{Z}}$ that take the form $\pi(a \mid s, z; s_0)$. *Effective empowerment* is the mutual information realized by a particular skill-conditioned policy at a particular starting state s_0 under the uniform skill source $p(z \mid s_0) = \text{Unif}(\mathcal{Z})$:

$$\begin{aligned} \mathcal{E}_{\text{eff}}(\pi, s_0) &= I^{\pi}(Z; S_+ \mid S_0 = s_0) \\ &= \mathbb{E}_{p(z \mid s_0)} D_{\text{KL}}[p^{\pi}(s_+ \mid s_0, z) \parallel p^{\pi}(s_+ \mid s_0)] \end{aligned}$$

This objective is similar to the objectives directly maximized in modern skill-learning algorithms [1, 5, 16]; note that MISL methods generally assume a fixed starting state s_0 , so the learned skills are specific to s_0 .

Importantly, effective empowerment involves no maximization and measures the MI *achieved* by a particular skill-conditioned policy $\pi \in \Pi_{\mathcal{Z}}$ at a particular starting state s_0 . This MI depends on both the *channel* $z \mapsto p^{\pi}(s_+ \mid s_0, z)$, determined by the policy, and the *source* $p(z \mid s_0)$ set to uniform. Potential empowerment is the maximum-achievable effective empowerment for a given starting state s_0 , so that

$$\mathcal{E}_{\text{pot}}(s_0) = \max_{\pi \in \Pi_{\mathcal{Z}}} \mathcal{E}_{\text{eff}}(\pi, s_0) \quad (6)$$

$$= \max_{\pi \in \Pi_{\mathcal{Z}}} \max_{p(z \mid s_0)} I^{\pi}(Z; S_+ \mid S_0 = s_0). \quad (7)$$

We can write the second equality because the source is overparameterized. Under a uniform skill prior, the policy can represent any skill source by simply relabeling skills [17]. Fixing the uniform source in Definition 3.1 therefore loses no generality.

Thus, a learned skill set (effective empowerment policies) acts as a probe of the potential empowerment at a starting state. If skills initialized at s_0 induce easily distinguishable future occupancies $p^{\pi}(s_+ \mid s_0, z)$, then s_0 affords many controllable futures and has high potential empowerment $\mathcal{E}_{\text{pot}}(s_0)$.

4 Methods

MISL methods learn skill-conditioned policies π_{eff} and value function that measure the *effective empowerment* from a state s_0 (Definition 3.3). These skills therefore serve as the yardsticks of empowerment: the diversity of these skills provides a lower bound on the potential empowerment of s_0 . A *potential empowerment* policy π_{pot} instead visits states with high potential empowerment:

$$\pi_{\text{pot}} \in \arg \max_{\pi \in \Pi} \mathbb{E}_{\pi} \left[\sum_t \gamma^t \mathcal{E}_{\text{pot}}(s_t) \right]. \quad (8)$$

This connection suggests a two-policy algorithm. The skill policy π_{eff} estimates a lower bound $\widehat{\mathcal{E}}_{\text{pot}}(s)$ on statewise potential empowerment, while the potential policy π_{pot} maximizes this estimate as an intrinsic reward. The effective and potential empowerment policies therefore support one another: the skills π_{eff} measure the controllable futures available from a state, while the potential empowerment policy π_{pot} moves the agent toward states from which the agent can exert control.

High-level method summary. The empowerment-maximization algorithm (Algorithm 1) alternates between training the two policies shown in Fig. 1. First, we train the skills π_{eff} and use their on-policy values to estimate a lower bound $\widehat{\mathcal{E}}_{\text{pot}}(s)$ on potential empowerment. We then use this estimate as an intrinsic reward for the potential policy π_{pot} . For our chosen MISL objective, the skill update is an effective-empowerment policy-improvement step under the assumptions of Theorem A.1, a modified version of the policy improvement results in [1]. Because this framework is general, arbitrary MISL estimators can take the place of the underlying skill-learning algorithm.

Algorithm 1: Empowerment maximization. Skills estimate the control afforded by each state, and π_{pot} ascends this estimate. Full version in Algorithm 2.

```

1: for  $n = 1, \dots, N$  do
2:   Skills: train  $\pi_{\text{eff}}$  with MISL
3:   Estimate:  $\widehat{\mathcal{E}}_{\text{pot}}(s) \leftarrow \mathbb{E}_{p(z)}[V_z(s)]$ 
4:   Ascend: train  $\pi_{\text{pot}}$  on reward  $\widehat{\mathcal{E}}_{\text{pot}}$ 
    
```

Training structure. Making the empowerment maximization recipe practical for high DoF settings raises key questions:

1. Which empowerment objective is appropriate to utilize modern USD methods?
2. How do we gather data to train the empowerment objective?
3. What are practical implementation details necessary to get empowerment maximization methods to work in high DoF settings?

4.1 USD to maximize Empowerment

4.2 How to gather data: skill switching

Switching policies. While we presented a methodology to convert skill-learning methods into potential empowerment maximization algorithms, we cannot simply learn skills for every starting state s_0 , bound the potential empowerment using the effective empowerment, then maximize the potential empowerment. For arbitrarily large state spaces, learning skills everywhere is exactly the type of intractable exploration problem that empowerment maximization wishes to avoid.

We consider a two-stage rollout scheme where we only consider effective empowerment maximization at reachable states. During data collection, we rollout both policies within a single trajectory.

The potential policy first moves the agent to a handoff state s_0 , after which the agent switches “on” the skills by sampling a skill z and executing $\pi_{\text{eff}}(\cdot \mid \cdot, z; s_0)$ (Fig. 1). The two phases serve distinct roles. The potential policy determines *where* the agent initiates skills, while the skill policy measures the controllable futures available from that state. Both of these behaviors are important. The potential policy enables the agent to modulate the starting state for the skills and estimate the empowerment *field*. Meanwhile, skills lower bound the empowerment of any fixed state. We can interpret skills as local probes of empowerment, where the potential empowerment policy ascends the empowerment landscape given by these probes. The two-phase rollout naturally constrains the skill learning to the reachable states and provides a useful training signal for the potential policy in large environments.

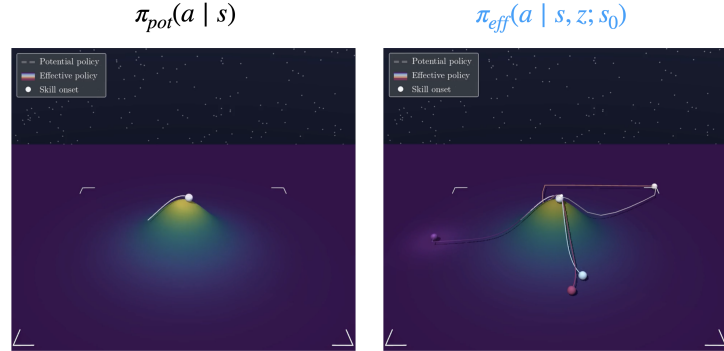


Figure 1: **Two trained empowerment-maximizing policies characterize central states in an MDP.** Consider the MDP of an agent (ball) on a hill, where the top represents a clearing. (Left) The potential empowerment policy (white) ascends the empowerment landscape, using the per-state empowerment $\mathcal{E}_{\text{pot}}(s)$ as an intrinsic reward. (Right) The effective empowerment policy π_{eff} (colors) maximizes $I^\pi(Z; S_+ \mid S_0)$ over skills z , and provides a lower bound on the potential empowerment.

4.3 How to learn policies: MISL

Contrastive Successor Features. We consider variants of Zheng et al. [16, Contrastive Successor Features (CSF)], a recent state-of-the-art skill-learning method, to lower-bound the effective empowerment $\mathcal{E}_{\text{eff}}(\pi) = I^\pi(Z; S_+ | S_0 = s_0)$ for a fixed starting state s_0 . CSF estimates the one-step MI $I(Z; S, S')$ from a contrastively-trained log density ratio; we instead train the classifier on transitions separated by a geometric random variable length of time $\tau \sim \text{Geom}(1 - \gamma)$, which gives the long-horizon MI conditioned on the handoff state. Fixing the skill prior to $p(z | s_0) \triangleq p(z) = \text{Unif}(\mathcal{Z})$ removes the otherwise intractable $p(z | s_0)$ from the estimator and loses no generality for continuous skill spaces [17]. The full derivation is in Appendix A.1.

Following CSF, we parametrize the log ratio with a bilinear form in a learned state encoder ϕ

$$f_\phi(z; s_0, s_+) = z^\top (\phi(s_+) - \phi(s_0)) \approx \log \frac{q_\phi(z | s_0, s_+)}{p(z)}, \quad (9)$$

measuring the displacement from the handoff state. Learning this log ratio involves training a classification loss utilizing InfoNCE [18] when $\mathcal{Z}$ is continuous or a multi-class cross-entropy loss over the M skills when $\mathcal{Z}$ is discrete.

Training the skills. Because the skill prior is uniform, Eq. (9) is precisely the reward of the effective empowerment objective:

$$r_n(s, a; z, s_0) \approx \mathbb{E}_{p(s' | s, a)} [f_\phi(z; s_0, s_+ = s')], \quad (10)$$

which, in expectation over discounted timesteps and skills, lower-bounds the effective empowerment $\mathcal{E}_{\text{eff}}(\pi)$. Under the assumptions of Theorem A.1, alternating the policy update and fitting f_ϕ on-policy improves the true MI.

Training the potential policy. Given an estimate of the skill value function V_z , improving the potential empowerment-seeking behavior reduces to standard reinforcement learning (RL) with an intrinsic reward given by the skill-based MI:

$$\widehat{\mathcal{E}}_{\text{pot}}(s) \approx \mathbb{E}_{p(z)} [V_z(s; s)] \approx I^{\pi_{\text{eff}}}(Z; S_+ | S_0 = s) \leq \mathcal{E}_{\text{pot}}(s). \quad (11)$$

so we train π_{pot} on $\widehat{\mathcal{E}}_{\text{pot}}$ with any standard RL algorithm.

5 Maximizing empowerment in high-dimensional, continuous settings

[CJ: Q: why the heck is SAC so much better for empowerment maximization in continuous settings? PPO is extremely unstable.]

The above results provide a route for potential empowerment maximization and theoretically-meaningful interpretations of empowerment via MISL. Practically, we build off of Zheng et al. [16, CSF] as the underlying MISL method, a recent state-of-the-art skill-learning method, to lower bound the effective empowerment $\mathcal{E}_{\text{eff}}(\pi) = I^\pi(S_+; Z | S_0 = s_0)$ for some fixed starting state s_0 . Further details on this specific MISL algorithm and pseudocode is in Appendix A. We briefly discuss empirical considerations for implementing the algorithm in practice and present continuous, high-DoF results. Training hyperparameters and detailed considerations are in Appendix B.

5.1 Practical considerations for empowerment maximization

Like the underlying USD methods, empowerment maximization has a few key practical considerations. We summarize the most important considerations below and provide theoretical intuition behind why these failure modes (and fixes) are inherent to the current parameterization of empowerment.

Increasing sample efficiency using hindsight relabeling The effective empowerment policy $\pi_{\text{eff}}(a | s; s_0, z)$ conditions on the current state, a skill, and a handoff state in $\mathcal{S} \times \mathcal{Z} \times \mathcal{S}$. This is a large space. To increase signal, we apply hindsight relabeling [19] during the effective empowerment policy and critic updates. Rather than the traditional HER of future goal state, we instead relabel s_0 by sampling backwards along the same trajectory, then assign the skill z with the highest classifier logit $f_\phi(z; s, s_+)$. Then, the policy can train on previously-visited states and handoff states under the current skill model, rather than strictly on data collected from the handoff state of the present rollout.

High MI $\neq$ interesting behavior. While empowerment is a useful metric to determine whether training has collapsed, high MI *does not necessarily imply interesting behavior*. Indeed, prior work has shown that small perturbations in continuous state space can lead to unbounded MI [17]. In other words, it is entirely possible to train a “successful” empowerment-maximization method that learns distinguishable skills with no *visually* distinguishable behavior, or behavior distinguishable to any reward function we may care about [CJ: add figures of this to put in app.] We exactly observe this collapse in behavior in DIAYN and VIC objectives.

The necessity of regularization. Thus, it is important to maximize MI subject to regularization. Prior works already identify this failure mode in unsupervised skill discovery literature. One possible solution is to compute (and maximize) empowerment over representations of state space, where the representations lie on a metric space. For example, [20] force representations to live on, approximately, a temporal distance space: the representations of two consecutive states in a trajectory are, on average, a unit of distance apart in metric space. If empowerment is a function of distances between representations, this constraint bounds (in expectation) the realizable empowerment in a single timestep, so only sustained behavior can induce high empowerment.

While the above solution incentivises learning sustained skills, this approach does not prevent sustained *collapsed* behavior. Indeed, applying METRA [20] alone to potential empowerment maximization leads to disordered, local perturbations while maximizing MI [figure]. Thus, beyond enforcing a metric structure, we need to noise important, observable degrees of freedom. Then, these degrees of freedom become harder to distinguish and require more dramatic perturbation to register as distinct skills. Importantly, the necessity of this regularization requires that the optimized effective empowerment *will not and should not* be the maximum possible MI.

Parameterization of MI. Some hypotheses to test:

- I think it’s p important to consider an asymmetric / true quasimetric for potential parameterization of MI. Unsure if this will work, will give a shot.

5.2 Results

Algorithm 2 learns π_{pot} and π_{eff} . Videos at this link; setups and hyperparameters in Appendix B.

- **Discrete, large.** Craftax: open-ended, partially observed, no task reward. [CJ: TODO: results, put these first.]
- **Continuous, high-DoF.** AntSoccer, BuilderBench: π_{pot} settles at a centralized state, π_{eff} fans out from centralized state, MI rises with no external reward (Figs. 2 and 3).
- **Choice of empowerment state space.** Full ant and ball state vs. ball alone: moves the centralized state (Appendices B.2.1 and B.2.2).

Baselines.

Empowerment over observation space versus state space. Skill-learning methods will often learn shortcuts to maximize empowerment. Given the opportunity to maximize empowerment by, e.g., turning on/off the lights rather than learning meaningful behavior, .

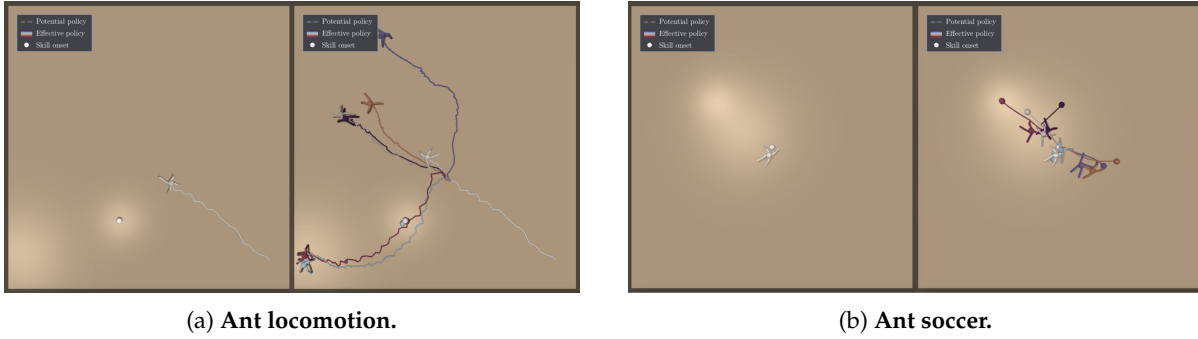


Figure 2: **Empowerment maximization on high-DoF Ant tasks.** (a) Ant locomotion: skills cover diverse movement directions, and the potential policy navigates to states with max directional diversity. (b) Ant soccer: skills kick ball in different directions, and the potential policy positions the agent for kicking and dribbling the ball.

Conditional MIs vs. Unconditioned MIs.

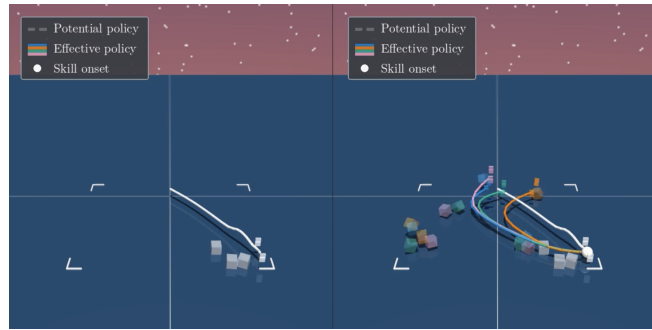


Figure 3: **Cubes combined.** Learned potential and effective empowerment policies in BuilderBench [21] with cubes initialized in the corner. (Left) The potential policy navigates the gripper behind the cubes and stays there, preparing to control the cubes. (Right) The effective policies perturb the cubes.

6 Discussion

Despite the conceptual appeal of empowerment as a tool for understanding natural agents and building artificial agents, its seeming computational intractability has caused prior work to resort to approximations that do not learn one policy that “keeps its options open” [13].

In providing a conceptual link between skill learning methods and empowerment, we provide a method that maximizes potential empowerment. We present key practical considerations to learn empowerment maximization policies in high-DoF settings and ablate various design decisions. Finally, we present the results of empowerment maximization in an open-ended setting, demonstrating that empowerment captures control-seeking behavior that is not found in other intrinsic reward methods. Future work should investigate simplified parameterizations of empowerment maximization that can cleanly resolve practical considerations detailed within our method, at scale.

Limitations.

References

- [1] Karol Gregor, Danilo Jimenez Rezende, and Daan Wierstra. *Variational Intrinsic Control*. *International Conference on Learning Representations*, 2016.
- [2] Shakir Mohamed and Danilo Jimenez Rezende. *Variational Information Maximisation for Intrinsically Motivated Reinforcement Learning*. *Neural Information Processing Systems*, 28, 2015.
- [3] A.S. Klyubin, D. Polani, and C.L. Nehaniv. *Empowerment: A Universal Agent-Centric Measure of Control*. *IEEE Congress on Evolutionary Computation*, volume 1, pp. 128–135 Vol.1, 2005.
- [4] Tobias Jung, Daniel Polani, and Peter Stone. *Empowerment for Continuous Agent—Environment Systems*. *Adaptive Behavior*, 19(1):16–39, 2011.
- [5] Benjamin Eysenbach, Abhishek Gupta, Julian Ibarz, and Sergey Levine. *Diver-*

- sity Is All You Need: Learning Skills Without a Reward Function. *International Conference on Learning Representations*, 2018.
- [6] Alison Gopnik. Empowerment as Causal Learning, Causal Learning as Empowerment: A Bridge Between Bayesian Causal Hypothesis Testing and Reinforcement Learning. 2024.
 - [7] Eunice Yiu, Kelsey Allen, Shiry Ginosar, and Alison Gopnik. Empowerment Gain and Causal Model Construction: Children and Adults Are Sensitive to Controllability and Variability in Their Causal Interventions. *arXiv:2512.08230*, 2026.
 - [8] Fei Dai and Alison Gopnik. Understand Intrinsic Motivation in Causal Learning Through Exploration. *Annual Meeting of the Cognitive Science Society*, volume 47, 2025.
 - [9] Aly Lidayan, Yuqing Du, Eliza Kosoy, Maria Rufova, Pieter Abbeel, and Alison Gopnik. Intrinsically-Motivated Humans and Agents in Open-World Exploration. *arXiv:2503.23631*, 2025.
 - [10] Claire Yang, Claire Jie Zhang, Maya Cakmak, and Max Kleiman-Weiner. When Assisting One Disempowers Another. *arXiv:2511.04177*, 2026.
 - [11] Yuqing Du, Stas Tiomkin, Emre Kiciman, Daniel Polani, Pieter Abbeel, and Anca Dragan. Ave: Assistance via Empowerment. *Neural Information Processing Systems*, 33:4560–4571, 2020.
 - [12] Andrew Levy, Alessandro G Allievi, and George Konidaris. Representation Learning and Skill Discovery With Empowerment. *Reinforcement Learning Conference*, 2025.
 - [13] Alexander S Klyubin, Daniel Polani, and Chrystopher L Nehaniv. Keep Your Options Open: An Information-Based Driving Principle for Sensorimotor Systems. *PloS One*, 3(12):e4018, 2008.
 - [14] Alexander S Klyubin, Daniel Polani, and Chrystopher L Nehaniv. All Else Being Equal Be Empowered. *European Conference on Artificial Life*, pp. 744–753, 2005.

- [15] Philippe Capdepuuy. Informational Principles of Perception-Action Loops and Collective Behaviours. 2011.
- [16] Chongyi Zheng, Jens Tuyls, Joanne Peng, and Benjamin Eysenbach. Can a MISL Fly? Analysis and Ingredients for Mutual Information Skill Learning. *ICLR 2025*, 2025.
- [17] Catherine Ji, Vivek Myers, Sergey Levine, and Benjamin Eysenbach. The Geometry of Empowerment. arXiv:TODO, 2026.
- [18] Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation Learning With Contrastive Predictive Coding. arXiv:1807.03748, 2018.
- [19] Marcin Andrychowicz, Filip Wolski, Alex Ray, Jonas Schneider, Rachel Fong, Peter Welinder, Bob McGrew, Josh Tobin, Pieter Abbeel, and Wojciech Zaremba. Hindsight Experience Replay. arXiv:1707.01495, 2018.
- [20] Seohong Park, Oleh Rybkin, and Sergey Levine. Metra: Scalable Unsupervised RL With Metric-Aware Abstraction. *International Conference on Learning Representations*, 2023.
- [21] Raj Ghugare, Roger Creus Castanyer, Catherine Ji, Kathryn Wantlin, Jin Schofield, Karthik Narasimhan, and Benjamin Eysenbach. BuilderBench: The Building Blocks of Intelligent Agents. arXiv:2510.06288, 2026.
- [22] David Barber and Felix Agakov. The IM Algorithm: A Variational Approach to Information Maximization. *Neural Information Processing Systems*, volume 16, 2003.
- [23] Martin L Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. John Wiley & Sons, 2014.
- [24] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft Actor-Critic: Off-Policy Maximum Entropy Deep Reinforcement Learning With a Stochastic Actor. *International Conference on Machine Learning*, pp. 1861–1870, 2018.
- [25] Alexander A. Alemi, Ian Fischer, Joshua V. Dillon, and Kevin Murphy.

Deep Variational Information Bottleneck. *International Conference on Learning Representations*, 2019.

A Algorithm Details

A.1 Training classifiers to optimize effective empowerment

We build on Zheng et al. [16, CSF], a recent state-of-the-art skill-learning method, to lower-bound the effective empowerment $\mathcal{E}_{\text{eff}}(\pi) = I^\pi(Z; S_+ | S_0 = s_0)$ for a fixed starting state s_0 .

The original CSF objective estimates the one-step mutual information $I(Z; S, S')$, where $s \sim p^\pi(s)$ is the marginal DSOM and s' is the next state. Let $q_\phi(z) = \text{Unif}(\mathcal{Z}) = p(z)$ and fix the starting state. Setting the one-step reward to the log ratio

$$r(s, a, s'; z) = \log \frac{q_\phi(z | s, s')}{p(z)} \quad (12)$$

yields an objective that is a variational lower bound on $I(Z; S, S')$ [16]:

$$\begin{aligned} \mathcal{O}_{\text{csf}}(\pi) &= \mathbb{E}_{p(z)} \sum_t \gamma^t r(s_t, a_t, s_{t+1}; z) \\ &= \mathbb{E}_{p^\pi(s, s', z)} \left[\log \frac{q_\phi(z | s, s')}{p(z)} \right] \\ &\leq I(Z; S, S'). \end{aligned} \quad (13)$$

Following CSF, the log-ratio estimator is trained with standard noise-contrastive objectives (e.g., InfoNCE) [16, 18]. The *conditional, long-horizon* MI, however, would require the different log ratio

$$\log \frac{q_\phi(z | s_0, s_+)}{q_\phi(z | s_0)}, \quad s_+ \sim p_z^\pi(s_+ | s_0), \quad z \sim p(z | s_0),$$

whose denominator $p(z | s_0)$ is generally intractable for high-dimensional z under standard contrastive estimation. This dependence turns out to be unnecessary: for continuous skill spaces, fixing $p(z | s_0) \triangleq p(z) = \text{Unif}(\mathcal{Z})$ and absorbing state-dependent skill selection into the policy loses no generality [17]. The uniform prior therefore removes the intractable $p(z | s_0)$ from the estimator entirely, so the long-horizon conditional MI becomes contrastively estimable. This is *not* generally true for discrete skill spaces, where a uniform prior over K skills restricts the source to the $1/K$ -quantized simplex [17].

The extension to long-horizon effective empowerment is then direct: rather than a classifier predicting z from one-step transitions, we train a classifier predicting z from transitions separated by a geometric random variable length of time $\tau \sim \text{Geom}(1 - \gamma)$. A sample of the one-step reward given (s_0, z, s') approximates

$$r(s, a; z, s_0) \approx \mathbb{E}_{p(s' | s, a)} \left[\log \frac{q_\phi(z | s_0, s_+ = s')}{p(z)} \right], \quad (14)$$

which, in expectation over discounted timesteps and skills, gives $\mathcal{O}(\pi) \leq I^\pi(Z; S_+ | S_0 = s_0) = \mathcal{E}_{\text{eff}}(\pi)$. This is a small modification to CSF (Eq. (13)), yielding a practical method for effective empowerment maximization.

A.2 Policy improvement for the MI that measures empowerment

The result below uses the standard variational information-maximization bound of Barber and Agakov [22], as used for skill discovery in Gregor et al. [1], together with classical policy improvement [23]. It shows that alternating policy improvement with classifier fitting is a valid policy-improvement scheme for the *true* MI objective, not merely its variational surrogate:

Theorem A.1 (Policy improvement for the MI that measures empowerment). *Let $0 < \gamma < 1$ be the discount factor and let $\pi_n(a | s, z; s_0)$ be the current skill-conditioned policy at iteration n , with $z \sim p(z)$ sampled at s_0*

and held fixed. Fit an on-policy classifier $q_n(z \mid s_0, s_+)$ to $p^{\pi_n}(z \mid s_0, s_+)$ and define the $(1 - \gamma)$ -normalized action value

$$Q_z^{\pi_n}(s, a; s_0) \triangleq (1 - \gamma) \mathbb{E}_{\pi_n(a \mid s, z; s_0)} \left[\sum_{k=0}^{\infty} \gamma^k \log \frac{q_n(z \mid s_0, s_{t+k+1})}{p(z)} \middle| s_t = s, a_t = a \right].$$

Let π_{n+1} be obtained by a greedy policy-improvement step with respect to $Q_z^{\pi_n}(s, a; s_0)$ for every z and s_0 . If the classifier is perfectly fit on-policy at each iteration, $q_n(z \mid s_0, s_+) = p^{\pi_n}(z \mid s_0, s_+)$, then

$$I^{\pi_{n+1}}(Z; S_+ \mid S_0) \geq I^{\pi_n}(Z; S_+ \mid S_0) \quad \forall n.$$

This is the theoretical license for the practical loop: even though the intrinsic reward is itself policy-dependent (via the classifier), each alternation monotonically improves the true empowerment objective, provided the classifier is refit on-policy. The proof is in Appendix C.1.

In practice, with an estimate of $V_z^{\pi_n}$, improving the *potential empowerment-seeking behavior* reduces to standard RL with intrinsic reward given by the skill-based MI:

$$r_n(s) \approx \mathbb{E}_{p(z)} [V_z^{\pi_n}(s; s)] \approx I^{\pi_n}(Z; S_+ \mid S_0 = s) \quad (15)$$

$$\leq \mathcal{E}_{\text{pot}}^n(s) = \max_{\pi \in \Pi_Z} I^\pi(Z; S_+ \mid S_0 = s), \quad (16)$$

which, under the γ -discounted sum, lower-bounds the cumulative potential empowerment objective:

$$\mathcal{O}_p^n(\pi_{\text{pot}}) \leq \mathbb{E}_{\pi_{\text{pot}}} \left[\sum_t \gamma^t \mathcal{E}_{\text{pot}}^n(s) \right].$$

The skill-conditioned effective empowerment policy $\pi_{\text{eff}}(a \mid s, z; s_0)$ and the potential policy $\pi_{\text{pot}}(a \mid s)$ are thus closely related but distinct.

A.3 Pseudocode

The discrete and continuous skill settings share the same loop, with f_ϕ parametrized as in Eq. (9), and differ only in how f_ϕ is fit: a multi-class cross-entropy loss over the M skills when $\mathcal{Z}$ is discrete, and a contrastive loss (InfoNCE [18], following CSF [16]) when $\mathcal{Z}$ is continuous. In both cases the policy update uses f_ϕ directly as the reward.

Algorithm 2: MISL for empowerment maximization

Require: MDP $\mathcal{M}$, discount γ , skill space $\mathcal{Z}$, skill prior $p(z) = \text{Unif}(\mathcal{Z})$, rollout length M , iterations N , buffer capacities $|\mathcal{B}_q| \ll |\mathcal{B}_\pi|$

- 1: **Initialize:** $\pi_{\text{eff}}(a \mid s; z, s_0)$, $\pi_{\text{pot}}(a \mid s)$, $f_\phi(z; s_0, s_+)$, $\mathcal{B}_q \leftarrow \emptyset$ (small), $\mathcal{B}_\pi \leftarrow \emptyset$ (large)
- 2: **for** $n = 1, \dots, N$ **do**
- 3: **Rollout:**
- 4: run π_{pot} for M steps, ending at handoff state s_0
- 5: sample skills $z_1, \dots, z_{K_z} \stackrel{\text{i.i.d.}}{\sim} p(z)$
- 6: **for** $k = 1, \dots, K_z$ **do**
- 7: initialize a skill rollout from the same handoff state s_0
- 8: run $\pi_{\text{eff}}(a \mid s, z_k, s_0)$ for αM steps
- 9: sample s_+ from the discounted future states of this skill rollout
- 10: append (s_0, z_k, s_+) to $\mathcal{B}_q$
- 11: append skill transitions $(s_t, a_t, s_{t+1}, s_0, z_k)$ to $\mathcal{B}_\pi$
- 12: **Classifier update:**
- 13: sample $(s_0, z, s_+) \sim p_{\mathcal{B}_q}$
- 14: update $f_\phi(z; s_0, s_+) \approx \log \frac{p(z|s_0, s_+)}{p(z)}$: cross-entropy if $\mathcal{Z}$ discrete, contrastive if $\mathcal{Z}$ continuous
- 15: **Effective-policy update:**
- 16: sample $s_+ \sim p_{\mathcal{B}_\pi}(s_+)$
- 17: HER sample the handoff state s_0 backwards in time from the same trajectory
- 18: assign $z \leftarrow \arg \max_{z' \in \mathcal{Z}} f_\phi(z'; s_0, s_+)$
- 19: update π_{eff} on $Q_z^{\pi_{\text{eff}}}(s, a; s_0)$ with reward $r_n = f_\phi(z; s_0, s_+)$
- 20: **Potential-policy update:**
- 21: sample minibatch $\sim \mathcal{B}_\pi$
- 22: update π_{pot} via SAC [24] with intrinsic reward $\mathcal{E}_{\text{pot}}^n(s) \approx \mathbb{E}_{p(z)}[V_z^{\pi_{\text{eff}}}(s; s)]$
- 23: **return** $\pi_{\text{eff}}, \pi_{\text{pot}}$

B Continuous Experiment Implementation Details

B.1 Practical considerations to scale potential empowerment maximization

Drawing skills from the uniform distribution. As described in Appendix A.1, we roll out skills from a marginal, *state un-conditioned* uniform distribution $z \sim p(z)$ rather than fitting the conditional source $p(z \mid s_0)$. This contrasts with Variational Intrinsic Control [1], which assumes and fits arbitrary source distributions $p(z \mid s_0)$. Furthermore, VIC does not simultaneously learn a potential empowerment policy, that is, a policy that maximizes empowerment as defined in [3].

Necessity of Information Bottlenecks information bottleneck (IB). In practice, maximizing the start state-conditioned MI in continuous settings leads to a failure mode: very small, almost imperceptible perturbations in state in, e.g., a deterministic settings can already saturate MI if using a constrained skill space. In fact, high MI *does not directly imply interesting behavior in continuous settings*.

Thus, it is critical to apply a form of information bottleneck from $S_0, S_+ \rightarrow Z$ when computing intrinsic rewards in the continuous settings. Adding Gaussian noise to states before feeding them into the classifier during training. This is a simplified case of applying the Variational Information Bottleneck [25]).

B.2 Hyperparameters and training results

B.2.1 AntSoccer, full obs

Here empowerment is defined over every degree of freedom, the coordinates of both the ant and the ball:

$$\mathcal{E}_{\text{eff}}(\pi) = I^\pi(Z; (S_{+, \text{ant}}, S_{+, \text{ball}}) \mid (S_{0, \text{ant}}, S_{0, \text{ball}})).$$

In this limit π_{pot} navigates to the center of the space and stays there (white, Fig. 2a), while the skills π_{eff} branch and cover the space (colors). The skills become more distinguishable because the agent first ascends the potential empowerment landscape. The estimated MI lower bound and the net agent displacement both grow over training, despite no external reward for locomotion.

Hyperparameters are in Table 1. Training curves are in Fig. 4a and Fig. 4b.

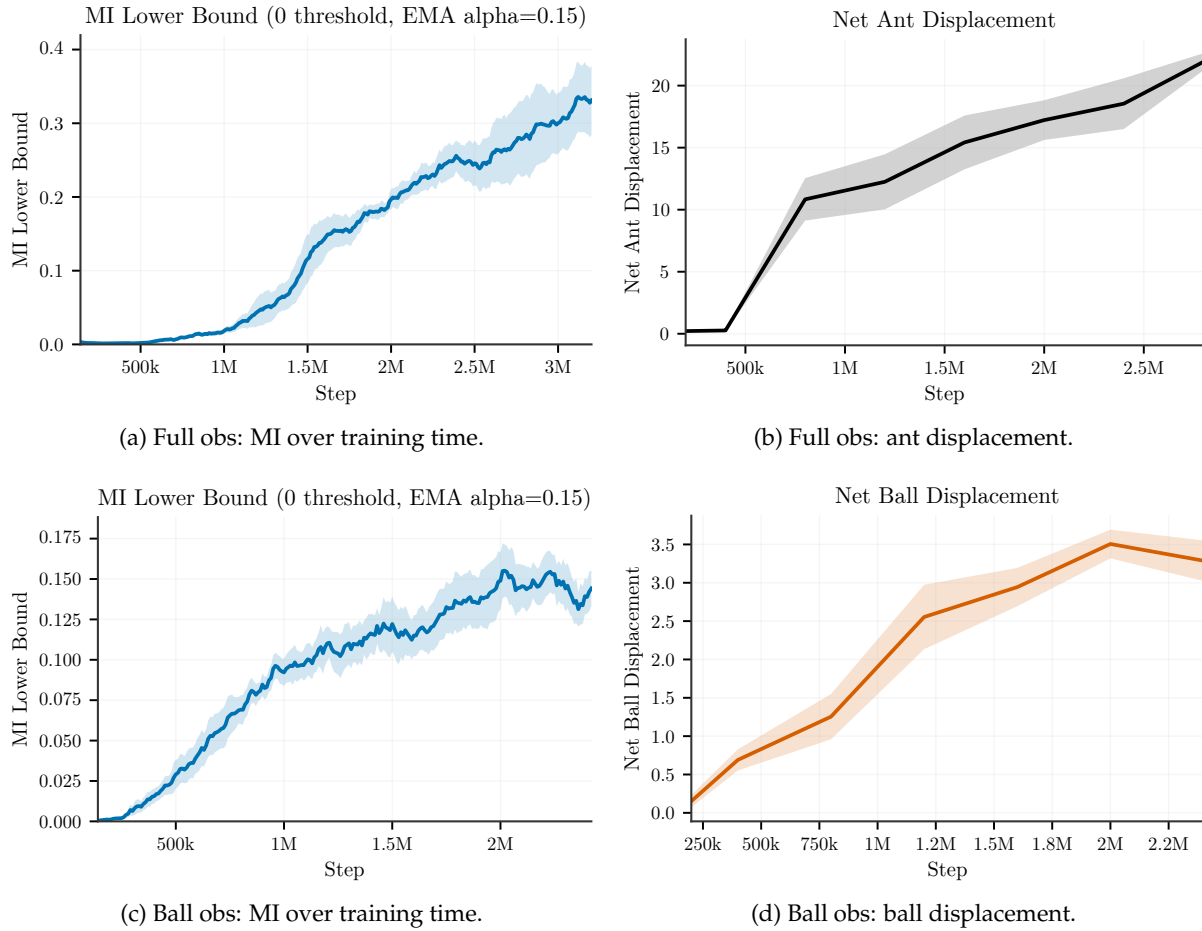


Figure 4: **AntSoccer training curves.** Full observations: (a) estimated MI lower bound and (b) ant displacement over training time. Ball observations only: (c) estimated MI lower bound and (d) ball displacement over training time. Over 5 seeds.

B.2.2 AntSoccer, ball obs only

Restricting the state space to the ball alone,

$$\mathcal{E}_{\text{eff}}(\pi) = I^\pi(Z; S_{+, \text{ball}} \mid S_{0, \text{ball}}),$$

Table 1: Hyperparameters in AntSoccer, full obs

Parameters	Value
State dimension	42
Action dimension	8
Skill dimension	2
Number of skills	30
Gamma for DSOM	0.9
Number of π_{pot} steps	100
Number of π_{eff} steps	300
# env steps	50_000
Update to env step ratio	1e-6
Vertex seed	0
Learning Rate	Linear interpolation from 1 $\rightarrow$ 0.01
Gamma for potential policy	0.9
# Q-learning iterations	2×10^5

the agent is empowered only if it learns to manipulate and control the ball. The centralized state moves accordingly: π_{pot} positions the ant to approach and kick the ball (Fig. 2b, left), and the skills π_{eff} then kick and dribble it in various directions (right).

Hyperparameters are in Table 2. Training curves are in Fig. 4c and Fig. 4d.

Table 2: Hyperparameters in AntSoccer, ball obs only

Parameters	Value
State dimension	42
Action dimension	8
Skill dimension	2
Number of skills	30
Gamma for DSOM	0.99
Number of π_{pot} steps	100
Number of π_{eff} steps	700
# env steps	3_000_000
Update to env step ratio	1:1
Learning Rate	3e-4
Gamma for policies	0.99

B.2.3 BuilderBench

BuilderBench [21] tests manipulation and open-endedness: a gripper manipulates three cubes initialized in the corner of a boxed play area (Fig. 3, white corners denote the play area). Defining empowerment over the cube xyz positions produces center-seeking behavior. The potential policy π_{pot} sneaks behind the cubes, where it has easy access to control them, and stays there. The skills π_{eff} then knock the cubes in different directions, and the MI rises over training as the arm learns to interact with them.

Hyperparameters are in Table 3. The figures of MI over environment steps and displacement over environment steps are in Fig. 5a and Fig. 5b.

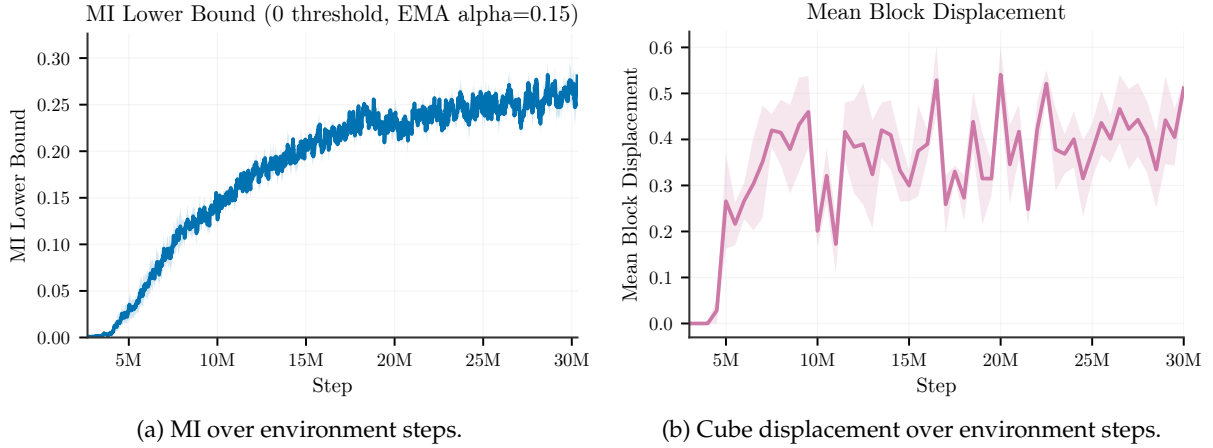


Figure 5: **BuilderBench with cube observations.** (a) Estimated MI lower bound and (b) cube displacement over environment steps. Over 4 seeds.

Table 3: Hyperparameters in BuilderBench

Parameters	Value
State dimension	51
Action dimension	5
Skill dimension	9
Number of skills	30
Gamma for DSOM	0.99
Number of π_{pot} steps	100
Number of π_{eff} steps	700
# env steps	30_000_000
Update to env step ratio	1:8
Learning Rate	3e-4
Gamma for policies	0.99

C Proofs

C.1 Policy improvement and MI lower bound

Theorem A.1 (Policy improvement for the MI that measures empowerment). *Let $0 < \gamma < 1$ be the discount factor and let $\pi_n(a \mid s, z; s_0)$ be the current skill-conditioned policy at iteration n , with $z \sim p(z)$ sampled at s_0 and held fixed. Fit an on-policy classifier $q_n(z \mid s_0, s_+)$ to $p^{\pi_n}(z \mid s_0, s_+)$ and define the $(1 - \gamma)$ -normalized action value*

$$Q_z^{\pi_n}(s, a; s_0) \triangleq (1 - \gamma) \mathbb{E}_{\pi_n(a \mid s, z; s_0)} \left[\sum_{k=0}^{\infty} \gamma^k \log \frac{q_n(z \mid s_0, s_{t+k+1})}{p(z)} \middle| s_t = s, a_t = a \right].$$

Let π_{n+1} be obtained by a greedy policy-improvement step with respect to $Q_z^{\pi_n}(s, a; s_0)$ for every z and s_0 . If the classifier is perfectly fit on-policy at each iteration, $q_n(z \mid s_0, s_+) = p^{\pi_n}(z \mid s_0, s_+)$, then

$$I^{\pi_{n+1}}(Z; S_+ \mid S_0) \geq I^{\pi_n}(Z; S_+ \mid S_0) \quad \forall n.$$

The proof separates two facts that are easy to conflate. Holding the classifier fixed gives an ordinary discounted MDP, so policy improvement applies. Refitting the classifier makes the variational MI bound tight again.

Proposition C.1 (A fixed classifier defines an ordinary intrinsic-reward MDP). *Fix a classifier $q(z \mid s_0, s_+)$ and define the normalized intrinsic reward*

$$r_q(s_0, z, s') \triangleq (1 - \gamma) \log \frac{q(z \mid s_0, s')}{p(z)}.$$

For any policy π , define

$$Q_z^{\pi, q}(s, a; s_0) \triangleq \mathbb{E}_{\pi(a \mid s, z; s_0)} \left[\sum_{k \geq 0} \gamma^k r_q(s_0, z, s_{t+k+1}) \mid s_t = s, a_t = a \right].$$

Then $Q_z^{\pi, q}$ satisfies the Bellman equation

$$\begin{aligned} Q_z^{\pi, q}(s, a; s_0) &= (1 - \gamma) \mathbb{E}_{p(s' \mid s, a)} \left[\log \frac{q(z \mid s_0, s')}{p(z)} \right] \\ &\quad + \gamma \mathbb{E}_{p(s' \mid s, a)} [V_z^{\pi, q}(s'; s_0)], \end{aligned}$$

where $V_z^{\pi, q}(s; s_0) = \mathbb{E}_{\pi(a \mid s, z; s_0)} Q_z^{\pi, q}(s, a; s_0)$.

Proof. Peel off the first term in the discounted sum and use the Markov property. The classifier is fixed, so r_q is a stationary reward for the induced MDP indexed by (z, s_0) . $\square$

Proposition C.2 (Barber–Agakov lower bound is tight at the on-policy posterior). *For any policy π and any classifier $q(z \mid s_0, s_+)$,*

$$\begin{aligned} \mathcal{L}(\pi, q) &\triangleq \mathbb{E}_{p(z) p_z^\pi(s_+ \mid s_0)} \left[\log \frac{q(z \mid s_0, s_+)}{p(z)} \right] \\ &\leq I^\pi(Z; S_+ \mid S_0 = s_0). \end{aligned}$$

Equality holds if $q(z \mid s_0, s_+) = p^\pi(z \mid s_0, s_+)$.

Proof. Write the MI as

$$I^\pi(Z; S_+ \mid S_0 = s_0) = H(Z) + \mathbb{E}_{p^\pi(s_+ \mid s_0)} \mathbb{E}_{p^\pi(z \mid s_0, s_+)} [\log p^\pi(z \mid s_0, s_+)].$$

For every s_+ ,

$$D_{\text{KL}}(p^\pi(z \mid s_0, s_+) \parallel q(z \mid s_0, s_+)) \geq 0,$$

so replacing $\log p^\pi$ by $\log q$ gives a lower bound. Since $H(Z) = -\mathbb{E}_{p(z)} \log p(z)$ and the joint law factors as $p(z) p_z^\pi(s_+ \mid s_0)$, the bound is exactly $\mathcal{L}(\pi, q)$. Equality holds at the on-policy posterior. $\square$

Proof of Theorem A.1. At iteration n , the classifier q_n is fit to the on-policy posterior, so

$$\mathcal{L}(\pi_n, q_n) = I^{\pi_n}(Z; S_+ \mid S_0).$$

Now hold q_n fixed. By Proposition C.1, this is an ordinary MDP with stationary intrinsic reward. A greedy policy-improvement step therefore gives

$$\mathcal{L}(\pi_{n+1}, q_n) \geq \mathcal{L}(\pi_n, q_n).$$

Finally refit the classifier to the new on-policy posterior. By Proposition C.2, the refitted classifier makes the lower bound tight and can only increase the lower-bound value relative to the old classifier:

$$\begin{aligned} I^{\pi_{n+1}}(Z; S_+ \mid S_0) &= \mathcal{L}(\pi_{n+1}, q_{n+1}) \\ &\geq \mathcal{L}(\pi_{n+1}, q_n) \\ &\geq \mathcal{L}(\pi_n, q_n) \\ &= I^{\pi_n}(Z; S_+ \mid S_0). \end{aligned}$$

Thus exact on-policy classifier fitting plus greedy improvement monotonically improves the true MI. $\square$

Glossary

$\mathcal{E}_{\text{eff}}$ the effective empowerment as a function of the policy . 5, 7, 14, 17

$\mathcal{E}_{\text{pot}}$ The potential empowerment at state s and iteration n , defined as the maximal possible mutual information between skills and future states under some policy. . 15, 16

$\mathcal{E}_{\text{pot}}$ the potential empowerment as a function of the state . 4–7, 15, 16, 21

π_{eff} an effective empowerment maximizing policy . 4–8, 15–19

π_{pot} a potential empowerment maximizing policy . 4–8, 15–19

Acronyms

CSF Contrastive Successor Features. 7, 14, 15

DSOM discounted state occupancy measure. 4, 14, 18, 19

IB information bottleneck. 16

MDP Markov Decision Process. 2, 4, 16

MI mutual information. 3, 5, 7–9, 14–19

MISL Mutual Information Skill Learning. 3–7, 16

RL reinforcement learning. 7, 15